

NUS QF3101: Investment Instrument and Risk Management

March 22, 2023

Computing Assignment

Instructor: Tan Hwee Huat

Student: Wira Azmoon Ahmad A0149286R

1 Three-month SOFR Futures and Deferred IRS

1.a Data Collection and Preparation

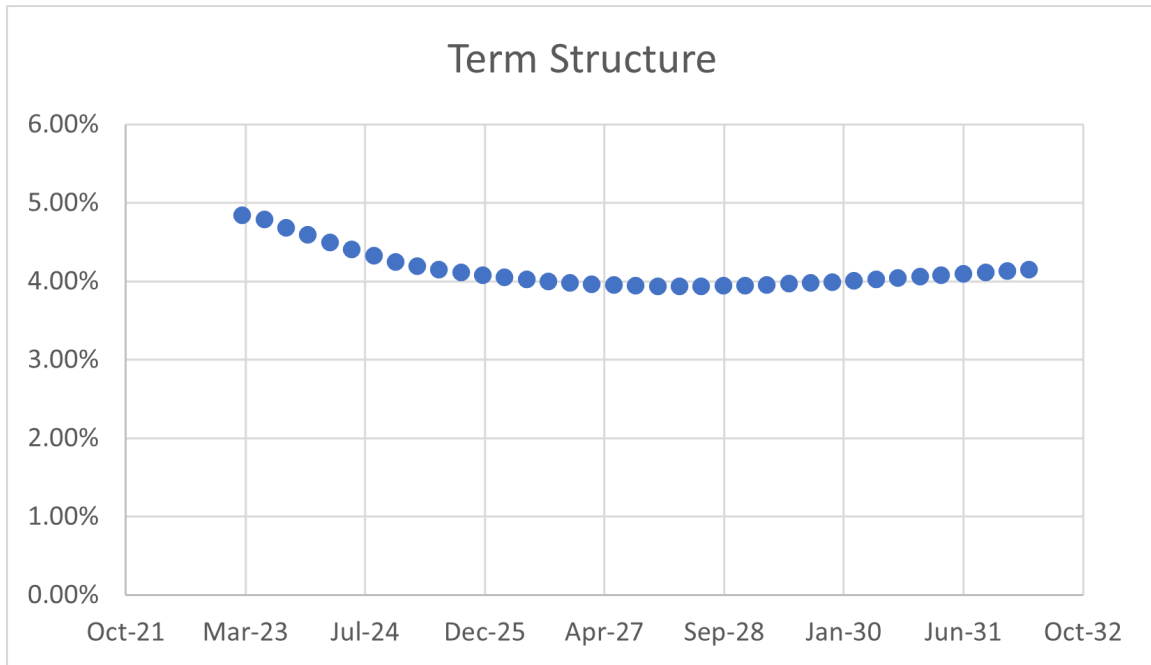
THREE-MONTH SOFR FUTURES - QUOTES

AUTO-REFRESH 10 SECS

Last Updated On Mar-2023 03:22 AM CDT

Market data is delivered by [Bloomberg](#) via

1.b Implied Forward SOFR Term Structure



The term structure is downward sloping for at least the next 4 years, which signifies that the economy is in, or about to enter, a recessive period. There is a slight uptrend afterwards till the end of the decade, signifying a predicted expansionary period.

1.c Swap Rate for Deferred Interest Rate Swap

K	1	2	3	4	5	6	7	8
Jun-23	4.36%	3.99%	3.80%	3.67%	3.61%	3.57%	3.56%	3.56%
Sep-23	4.12%	3.82%	3.68%	3.57%	3.53%	3.51%	3.51%	3.51%
Dec-23	3.91%	3.70%	3.58%	3.50%	3.47%	3.47%	3.47%	3.48%
Mar-24	3.73%	3.59%	3.49%	3.44%	3.43%	3.43%	3.44%	3.46%

Similar to the term structure, the swap rate drops for longer swap periods. It is also lower for further deferred start periods compared to earlier.

2 Efficient Equity Portfolios

2.a Data collection and preparation.

Average rate of return

$$r = \begin{bmatrix} 0.5583\% & 0.1963\% & 0.1510\% & 0.2355\% & -0.0618\% \end{bmatrix}^T$$

Covariance matrix

$$C = \begin{bmatrix} 0.00176 & 0.00064 & 0.00061 & 0.00099 & 0.00074 \\ 0.00064 & 0.00094 & 0.00070 & 0.00111 & 0.00080 \\ 0.00061 & 0.00070 & 0.00179 & 0.00100 & 0.00075 \\ 0.00099 & 0.00111 & 0.00100 & 0.00392 & 0.00179 \\ 0.00074 & 0.00080 & 0.00075 & 0.00179 & 0.00297 \end{bmatrix}$$

On average, AAPL performs best, while HAS performs worst. The covariance matrix is entirely positive, so it indicates that all 5 stocks tend to move in the same direction.

2.b Minimum Variance Point Portfolio And An Efficient Portfolio.

Minimum Variance portfolio with weight vector

$$w_G = \begin{bmatrix} 0.19281 & 0.69828 & 0.15050 & -0.10442 & 0.06284 \end{bmatrix}^T$$

with expected rate of return

$$r_G = 0.2390\%$$

and variance 0.000818.

An efficient portfolio with weight vector

$$w_m = \begin{bmatrix} 1.159769 & 0.311124 & -0.061948 & 0.049304 & -0.458249 \end{bmatrix}^T$$

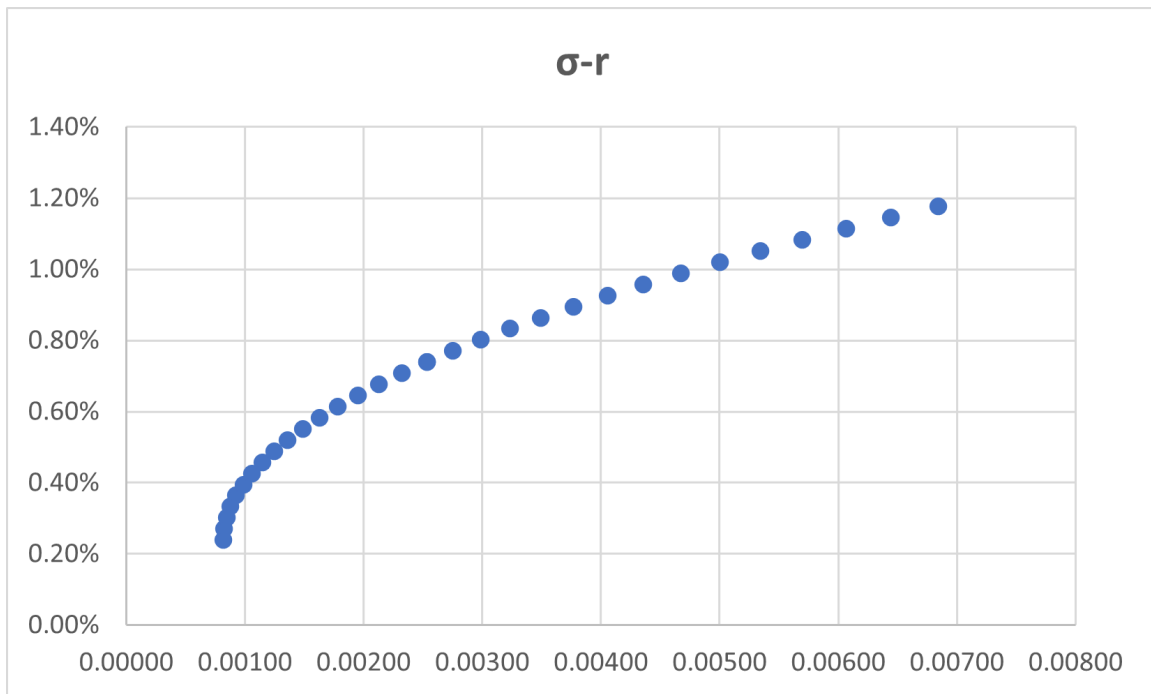
with expected rate of return

$$r_m = 0.7392\%$$

and variance 0.002531.

It is clear that any portfolio other than the minimum variance portfolio will have a higher variance.

2.c Efficient Portfolios



Using the Two-Fund Theorem, we can choose any portfolio with weights

$$w = \alpha w_m + (1 - \alpha)w_G$$

where $\alpha > 0$ to have an efficient portfolio with return $r > r_G$. These portfolios are on the efficient frontier, visible on the $\sigma - \bar{r}$ graph above.